

Mount Kenya University

Unlocking Infinite Possibilities

SCHOOL OF COMPUTING AND INFORMATICS

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

UNITE TITLE: ADVANCED WEB DESIGN AND DEVELOPMENT

UNITE CODE: BIT3208

PROJECT TITLE: MOONLIGHT ENTERPRISES

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Table of Contents

1.0 INSTALLATION LOCALHOST TOOL	4
1.1 XAMPP	4
1.2 Tested localhost successfully	4
1.3 Created Hello World pages	4
1.4 Tested basic database connectivity	5
2.0 Website Design and Project Planning	5
2.1 Designed Wireframes.....	5
2.2 Designed GUI Layouts	5
2.3 Planned Project Workflow	6
2.4 Selected Technologies.....	6
3.0 MOONLIGHT POS SYSTEM REPORT.....	6
3.1 Login Page	7
3.2 Dashboard.....	7
4.0 INVENTORY MANAGEMENT(CRUD)	7
4.1 Stock Inventory Page	8
4.2 CRUD OPERATIONS	8
4.3 Add Item.....	8
4.4 Delete function	8
4.5 Update.....	9
5.0 SALES MANAGEMENT	9
5.1 Record New Sale	9
5.2 Sales Records	9
6.0 DATABASE AND GITHUB	10
6.1 Database Integration	10
6.2 phpMyAdmin Database Table.....	10
6.3 Database connection code.....	11
6.4 GitHub Version Control.....	11
6.5 GitHub Repository System	11
7.0 CONCLUSION.....	11
8.0 REFERENCE.....	12

LIST OF FIGURES

Figure 1:XAMPP INSTALLATION	4
Figure 2:LOCALHOST TESTING	4
Figure 3:CODE	4
Figure 4:CREATION OF HELLO WORLD.....	5
Figure 5:TESTING DATA CONNECTION.....	5
Figure 6:DESIGNED FRAME	5
Figure 7:Designed GUI Layouts	6
Figure 8:Planned Project Workflow	6
Figure 9:LOGIN PAGE	7
Figure 10:DASHBOARD.....	7
Figure 11:Stock Inventory Page	8
Figure 12:ADD ITEM.....	8
Figure 13:DELETE FUNCTION	8
Figure 14:UPDATE	9
Figure 15:Record New Sale	9
Figure 16:Sales Records	10
Figure 17:phpMyAdmin Database Table	10
Figure 18:Database connection code	11
Figure 19:GitHub Repository System	11

1.0 INSTALLATION LOCALHOST TOOL

1.1 XAMPP

XAMPP was installed successfully to create a local server environment for web development. Apache and MySQL services were included to help run websites and manage databases on the computer.



Figure 1:XAMPP INSTALLATION

1.2 Tested localhost successfully

The localhost was tested by opening `http://localhost` in the browser. The XAMPP dashboard appeared successfully, showing that the Apache server was running correctly and the local environment was ready for web development.



Figure 2:LOCALHOST TESTING

1.3 Created Hello World pages



Figure 3:CODE



Figure 4: CREATION OF HELLO WORLD

1.4 Tested basic database connectivity

A database connection test was performed using PHP and MySQL. The successful connection message confirmed that the web application could communicate properly with the database server.

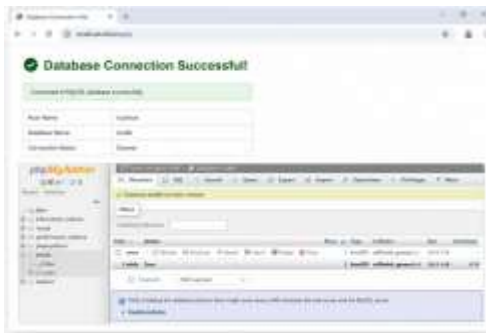


Figure 5: TESTING DATA CONNECTION

REFLECTION

I successfully installed XAMPP and configured the localhost environment for web development. This helped me understand how local servers work and how to run web applications on my computer.

2.0 Website Design and Project Planning

2.1 Designed Wireframes

Basic sketches were created to show the structure and layout of the website before development.



Figure 6: DESIGNED FRAME

2.2 Designed GUI Layouts

User interface designs were created to show how the web pages would look and function.



Figure 7:Designed GUI Layouts

2.3 Planned Project Workflow

The steps and activities of the project were organized to ensure smooth development.

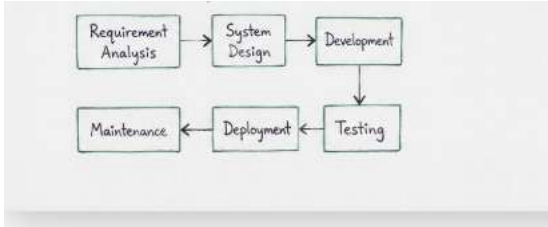


Figure 8:Planned Project Workflow

2.4 Selected Technologies

Appropriate tools and technologies such as HTML, CSS, PHP, MySQL, and XAMPP were chosen for the project development.

REFLECTION

I designed wireframes, GUI layouts, and planned the workflow of the project before development. This improved my skills in organizing and planning a complete web application project.

3.0 MOONLIGHT POS SYSTEM REPORT

Moonlight POS is a web-based point of sale and Inventory Management System developed using PHP, MySQL, HTML, CSS and JavaScript. The system manages inventory, sales records and transaction tracking.

3.1 Login Page



Figure 9:LOGIN PAGE

3.2 Dashboard



Figure 10:DASHBOARD

REFLECTION

I developed the Moonlight POS system with a login page and dashboard using PHP, HTML, CSS, JavaScript, and MySQL. This helped me gain practical experience in creating a functional web-based system.

4.0 INVENTORY MANAGEMENT(CRUD)

Inventory Module

The inventory section stores and manages product information such as item name, stock code, quantity, supplier and pricing details.

4.1 Stock Inventory Page



ID	Item Name	Stock Code	Quantity	Purchase Price	Selling Price	Supplier	Date Received	Received By	Actions
21	Sabred	QABR00	300	Ksh 200.00	Ksh 280.00	Aburadash	2025-10-11	Ido	Edit Delete
18	Mono SD 4GB	QMSD01	145	Ksh 4.00	Ksh 13.00	MonogPro	2025-10-17	Ido	Edit Delete
15	HERO 1000 2m	HER1000	75	Ksh 5.00	Ksh 12.00	HeroShop	2025-10-17	Ido	Edit Delete
22	Head	HS44	65	Ksh 300.00	Ksh 500.00	Ido	2025-10-11	@monono	Edit Delete
27	USB Cable	USBC00	45	Ksh 10.00	Ksh 20.00	MonogPro	2025-10-18	Ido	Edit Delete
14	Polar Book 100	POLB0010	40	Ksh 25.00	Ksh 40.00	MonogPro	2025-10-17	Ido	Edit Delete
25	Tablet PC	TSB001	65	Ksh 4,000.00	Ksh 5,000.00	Ido	2025-10-18	Ido	Edit Delete

Figure 11: Stock Inventory Page

4.2 CRUD OPERATIONS

The system implements CRUD operations:

1. Create -add new products.
2. Read -view inventory records.
3. Update -edit product information.
4. Delete -remove unwanted records.

4.3 Add Item



➕ Add New Item to Inventory

Use this form to add a brand new product to your stock list.

Item Name:

Unique Stock Code (SKU):

Initial Stock Quantity:

Purchase Price (Cost):

Selling Price (Retail):

Supplier:

Figure 12: ADD ITEM

4.4 Delete function



Current Stock Inventory

Deleted view of all items, quantities, and pricing. Items with quantity less than or equal to 0 are flagged.

Error! Cannot delete item ID 21. It has 4 associated sales records! Please delete the associated sales records first.

ID	Item Name	Stock Code	Quantity	Purchase Price	Selling Price	Supplier	Date Received	Received By	Actions
21	Sabred	QABR00	300	Ksh 200.00	Ksh 280.00	Aburadash	2025-10-11	Ido	Edit Delete
18	Mono SD 4GB	QMSD01	145	Ksh 4.00	Ksh 13.00	MonogPro	2025-10-17	Ido	Edit Delete
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22	Head	HS44	65	Ksh 300.00	Ksh 500.00	Ido	2025-10-11	@monono	Edit Delete
27	USB Cable	USBC00	45	Ksh 10.00	Ksh 20.00	MonogPro	2025-10-18	Ido	Edit Delete

Figure 13: DELETE FUNCTION

4.5 Update



Figure 14:UPDATE

REFLECTION

I implemented CRUD operations to add, view, update, and delete inventory records. Through this, I learned how to manage database records dynamically in a web application.

5.0 SALES MANAGEMENT

5.1 Record New Sale

The sales module allows users to record complete sales transactions into the system database.



Figure 15:Record New Sale

5.2 Sales Records

The system displays transaction history including quantity sold, sale amount, date and salesperson information.

Moonlight POS

Dashboard

Stock Inventory

All Item Item

Sales Records

Recent Sales Note

Sales History and Transaction Records

Detailed log of all completed sales transactions

Sub ID	Item Name	Stock Code	Qty Sold	Unit Price (per Item)	Total Sale	Date & Time	Sold By
01	Item A	ITEM001	5	\$180.000.000	\$900.000.000	2024-05-20 12:34	John
02	Item A	ITEM001	3	\$180.000.000	\$540.000.000	2024-05-20 14:56	John
03	Item B	ITEM002	2	\$250.000.000	\$500.000.000	2024-05-21 09:18	John
04	Item B	ITEM002	1	\$250.000.000	\$250.000.000	2024-05-21 11:34	John
05	Item C	ITEM003	4	\$120.000.000	\$480.000.000	2024-05-22 10:15	John
06	Item C	ITEM003	2	\$120.000.000	\$240.000.000	2024-05-22 13:45	John
07	Item D	ITEM004	3	\$160.000.000	\$480.000.000	2024-05-23 08:22	John
08	Item D	ITEM004	1	\$160.000.000	\$160.000.000	2024-05-23 11:54	John
09	Item E	ITEM005	2	\$180.000.000	\$360.000.000	2024-05-24 09:10	John
10	Item E	ITEM005	1	\$180.000.000	\$180.000.000	2024-05-24 13:22	John
11	Item F	ITEM006	3	\$150.000.000	\$450.000.000	2024-05-25 10:05	John
12	Item F	ITEM006	1	\$150.000.000	\$150.000.000	2024-05-25 14:30	John
13	Item G	ITEM007	2	\$170.000.000	\$340.000.000	2024-05-26 09:45	John
14	Item G	ITEM007	1	\$170.000.000	\$170.000.000	2024-05-26 13:12	John
15	Item H	ITEM008	4	\$140.000.000	\$560.000.000	2024-05-27 11:00	John
16	Item H	ITEM008	2	\$140.000.000	\$280.000.000	2024-05-27 15:22	John

Figure 16: Sales Records

REFLECTION

I created features for recording sales and viewing sales records. This improved my understanding of transaction management and data handling within the system.

6.0 DATABASE AND GITHUB

6.1 Database Integration

The system uses PHP and MySQL database connection to store, retrieve, update and manage data dynamically.

6.2 phpMyAdmin Database Table

Figure 17:phpMyAdmin Database Table

6.3 Database connection code

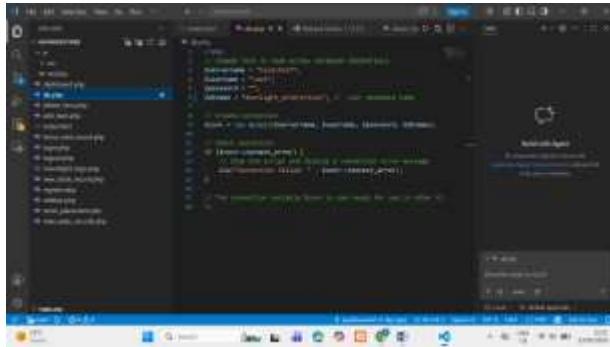


Figure 18:Database connection code

6.4 GitHub Version Control

GitHub was used to save project progress, maintain backups and track development updates.

6.5 GitHub Repository System

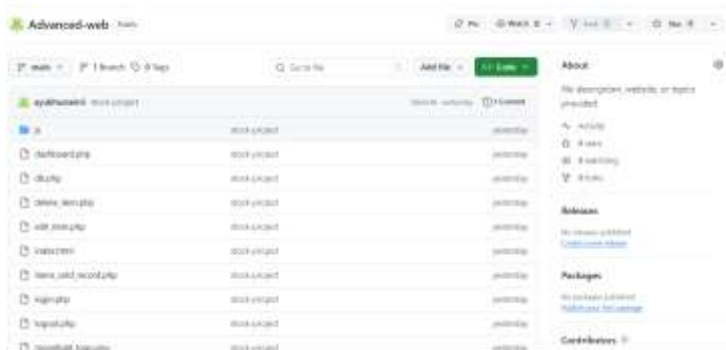


Figure 19:GitHub Repository System

<https://github.com/ayubhussein5/Advanced-web>

REFLECTION

I connected the system to a MySQL database and used GitHub for version control and backups. This helped me understand database integration and the importance of managing project versions online.

7.0 CONCLUSION

Moonlight POS successfully demonstrates backend development, database integration, authentication, inventory management, sales processing and CRUD operations.

8.0 REFERENCE

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